

UMP Unbiased Test

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Para Inf

UMP test may not exist. Ex: $X_1, \dots, X_n \stackrel{iid}{\sim} \mathcal{N}(\mu, \sigma_0^2)$ $\xrightarrow{\text{known}}$

$$\begin{aligned} \text{Test: } H_0: \mu = \mu_0 \text{ vs. } H_1': \mu > \mu_0 & \quad \phi_1(x) = \mathbb{1}\left(\frac{\sqrt{n}(\bar{x} - \mu_0)}{\sigma_0} > z_\alpha\right) \\ H_0: \mu = \mu_0 \text{ vs. } H_1': \mu < \mu_0 & \quad \phi_2(x) = \mathbb{1}\left(\frac{\sqrt{n}(\bar{x} - \mu_0)}{\sigma_0} < z_{1-\alpha}\right) \end{aligned}$$

$$\beta\phi_1(\mu) = 1 - \Phi\left(z_\alpha - \frac{\sqrt{n}(\mu - \mu_0)}{\sigma_0}\right) \uparrow \mu \in \mathbb{R}$$

$$\beta\phi_2(\mu) = \Phi\left(z_{1-\alpha} - \frac{\sqrt{n}(\mu - \mu_0)}{\sigma_0}\right) \downarrow \mu \in \mathbb{R}$$

Restrict our search for UMP in the class of unbiased tests

Will show: $\phi^* = \mathbb{1}\left(\frac{\sqrt{n}|\bar{x} - \mu_0|}{\sigma_0} > z_{\alpha/2}\right)$. UMP unbiased

size- α

Set up: Suppose X has pmf/pdf $f(x|\theta)$; $\theta \in \Theta$

Consider testing $H_0: \theta \in \Theta_0$ vs. $\theta \in \Theta_1$; where

$$\Theta_0 \cap \Theta_1 = \emptyset$$

$$\Theta_0 \cup \Theta_1 = \Theta$$

Defⁿ: A test ϕ is unbiased level- α (for testing H_0 vs. H_1)

$$\text{if } \beta\phi(\theta) \leq \alpha \quad \forall \theta \in \Theta_0 \text{ and } \beta\phi(\theta) \geq \alpha \quad \forall \theta \in \Theta_1$$

This means ϕ is at least as good as the silly test
 $\phi \equiv \alpha$

Defⁿ: A test ϕ^* (for taking H_0 vs. H_1) is called uniformly most powerful unbiased (UMPU) level- α test if

$$(i) \quad \beta\phi^*(\theta) \leq \alpha \quad \forall \theta \in \Theta_0 \text{ (level-}\alpha\text{)}$$

$$(ii) \quad \beta\phi^*(\theta) \geq \alpha \quad \forall \theta \in \Theta_1 \text{ (unbiased)}$$

$$(iii) \quad \beta\phi^*(\theta) \geq \beta\phi(\theta) \quad \forall \theta \in \Theta_1 \text{ for any } \phi \text{ satisfying (i) and (ii)}$$

$\S_0$ UMPU level- α test is the UMP test within class of unbiased level- α tests

Need to consider tests based on sufficiency statistic

Defⁿ: Let $\Theta_B := \overline{\Theta}_0 \cap \overline{\Theta}_1 =$ common boundary of Θ_0 and Θ_1 . A test ϕ is called similar on the boundary Θ_B (or α -similar) (SOB) if $\beta\phi(\theta) = \alpha \quad \forall \theta \in \Theta_B$

Lemma: If ϕ is unbiased level- α and $\beta\phi(\theta)$ is a continuous fn. of θ then ϕ is SOB on Θ_B i.e., $\beta\phi(\theta) = \alpha \quad \forall \theta \in \Theta_B$

Pf: Suppose $\theta_0 \in \Theta_B$. Consider $\{\theta_n\} \subseteq \Theta_0$ s.t. $\theta_n \rightarrow \theta_0$ then $\beta\phi(\theta_0) = \lim_{n \rightarrow \infty} \beta\phi(\theta_n) \leq \alpha$. If $\{\theta_n\} \subseteq \Theta_1$ s.t. $\theta_n \rightarrow \theta_0$ then $\beta\phi(\theta_0) = \lim_{n \rightarrow \infty} \beta\phi(\theta_n) \geq \alpha$ ($\because$ unbiasedness)

$$\therefore \beta\phi(\theta_0) = \alpha \quad \forall \theta_0 \in \Theta_B$$

Remark: If $\beta\phi(\theta)$ cont in θ for any test ϕ

$$\mathcal{U} = \{\text{unbiased level-}\alpha \text{ tests}\} \subseteq \{\text{SOB level-}\alpha \text{ tests}\} = \mathcal{J}$$

Lemma: Suppose $f(\alpha|\theta)$ is such that the power function $\beta\phi(\theta)$ is continuous for any test ϕ . If ϕ_0 is UMP test within the class of all SOB tests and ϕ_0 is level- α then ϕ_0 is UMPU level- α test.

Pf: We know $\mathcal{J} \supseteq \mathcal{U}$: ϕ_0 is as powerful as any test in $\mathcal{U}$

T.S. $\phi_0 \in \mathcal{U}$, consider $\psi \equiv \alpha$ and note $\psi \in \mathcal{S}$
 $\beta_\psi(\theta) = \alpha \forall \theta \in \Theta_B$

power of ψ , $\beta_\psi(\theta) = \alpha \leq \beta_{\phi_0}(\theta) \forall \theta \in \Theta_1 \Rightarrow \phi_0 \in \mathcal{U}$

$\therefore \phi_0$ is UMPU level- α

Recall: Suppose $f(x|\theta) = h(x)c(\theta)e^{\sum_{i=1}^k \theta_i T_i(x)}$; $\theta = (\theta_1, \dots, \theta_k)$
 (Canonical form)

If $g(x)$ is an integrable f.r. st

$\psi(\theta) = E_\theta g(X) = \int g(x)f(x|\theta)dx$ is finite $\forall \theta \in \Theta$

then $\psi(\theta)$ has derivative of all orders and $\frac{\partial}{\partial \theta_i}$ and $\int$ can be interchanged.

$\beta_\phi(\theta) = \int \phi(x)f(x|\theta) \rightarrow$ is continuous

UMPU test for two-sided hypothesis in one-parameter exp. family

Suppose X has pmf/pdf $f(x|\theta) = h(x)c(\theta)e^{\theta T(x)}$; $\theta \in \Theta \subset \mathbb{R}$

Already considered: UMP test for 1) $H_0: \theta \leq \theta_0$ vs. $H_1: \theta > \theta_0$
 2) $H_0: \theta \leq \theta_1$ or $\theta \geq \theta_2$

Theorem: (a) For testing (1) $H_0: \theta_1 \leq \theta \leq \theta_2$ vs. $H_1: \theta < \theta_1$ or $\theta > \theta_2$
 a UMPU size- α test is given by:

$$\phi^*(x) = \psi^*(T(x)) = \begin{cases} 1 & \text{if } T(x) < c_1 \text{ or } > c_2 \\ \gamma_i & \text{if } T(x) = c_i \text{ for } i=1,2 \\ 0 & \text{o.w.} \end{cases}$$

where $c_1, c_2, \gamma_1, \gamma_2$ are determined

by $E_{\theta_1} \phi^*(X) = E_{\theta_2} \phi^*(X) = \alpha$

(b) For testing $H_0: \theta = \theta_0$ vs. $H_1: \theta \neq \theta_0$, a UMPU size- α test

is given by

$$\phi^*(x) = \psi^*(T(x)) = \begin{cases} 1 & \text{if } T(x) < c_1 \text{ or } > c_2 \\ \gamma_i & \text{if } T(x) = c_i \text{ for } i=1,2 \\ 0 & \text{o/w} \end{cases}$$

where $c_1, c_2, \gamma_1, \gamma_2$ are determined by

$$(i) E_{\theta_0}(\psi^*(T)) = \alpha$$

$$(ii) E_{\theta_0}(T\psi^*(T)) = \alpha E_{\theta_0} T$$

Further if T is symmetrically distributed around point a under θ_0 then $\begin{cases} c_2 = 2a - c_1, \gamma_1 = \gamma_2, P_{\theta_0}(T=c_1) = P_{\theta_0}(T=c_2) \\ \text{solve for } c_1, \gamma_1 : P_{\theta_0}(T < c_1) + \gamma_1 P_{\theta_0}(T=c_1) = \alpha/2 \end{cases}$

Pf Sketch: Consider $\psi(T)$ based on sufficient stat only

note $\beta_\psi(\theta)$ is differentiable in θ

For any unbiased test ψ , we have (i) SOB: $\beta_\psi(\theta_0) = \alpha$

(ii) $\beta_\psi(\theta)$ is min at $\theta = \theta_0$ ($\because \beta_\psi(\theta) \geq \alpha \forall \theta \neq \theta_0$)

Note: Also $\beta'_\psi(\theta_0) = 0$

$$f_T(t|\theta) = h_1(t) c(\theta) e^{\theta t}$$

$$\beta'_\psi(\theta) = \frac{\partial}{\partial \theta} \int \psi(t) h_1(t) c(\theta) e^{\theta t} dt$$

$$\beta_\psi(\theta) = \int \psi(t) f_T(t|\theta) dt$$

$$= \int \psi(t) h_1(t) \underbrace{\frac{\partial}{\partial \theta} [c(\theta) e^{\theta t}]}_{c'(\theta) e^{\theta t} + c(\theta) t e^{\theta t}} dt$$

$$= \frac{c'(\theta)}{c(\theta)} \int \psi(t) \underbrace{h_1(t) c(\theta) e^{\theta t}}_{f_1(t)} dt + \int \psi(t) t \underbrace{h_1(t) c(\theta) e^{\theta t}}_{f_1(t)} dt$$

$$= \frac{c'(\theta)}{c(\theta)} E_\theta \psi(T) + E_\theta (T\psi(T)) \dots *$$

The trivial test $\psi_0 \equiv \alpha$ is unbiased, hence $(*)$ holds for ψ_0

$$\Rightarrow \beta'_{\psi_0}(\theta) = 0 = \frac{c'(\theta)}{c(\theta)} \alpha + \alpha \mathbb{E}_{\theta} T$$

$$\frac{c'(\theta)}{c(\theta)} = -\mathbb{E}_{\theta} T$$

$$(*) \Rightarrow \beta'_{\psi}(\theta) = -\mathbb{E}_{\theta} T \cdot \mathbb{E}_{\theta} \psi(T) + \mathbb{E}_{\theta} (T \psi(T))$$

$$\therefore \beta'_{\psi}(\theta_0) = 0 \Rightarrow \mathbb{E}_{\theta_0} (T \psi(T)) = \mathbb{E}_{\theta_0} T \mathbb{E}_{\theta_0} \psi(T) = \alpha \mathbb{E}_{\theta_0} T \quad (\text{and } \mathbb{E}_{\theta_0} \psi(T) = \alpha)$$

$$\mathcal{U} = \{\text{unbiased size-}\alpha \text{ test}\} \subseteq \{\text{tests satisfying (i), (ii)}\} = \mathcal{B}$$

Fix $\theta' \neq \theta_0$ Problem: To maximize $\int \psi(t) f_T(t|\theta') dt = \beta_{\psi}(\theta')$
subject to (i) $\int \psi(t) \underbrace{f_T(t|\theta_0)}_{f_1} dt = \alpha$ (ii) $\int \psi(t) t \underbrace{f_T(t|\theta_0)}_{f_2} dt = \alpha \mathbb{E}_{\theta_0} T$

Assuming $(\alpha, \alpha \mathbb{E}_{\theta_0} T)$ interior of $M = \{(\int \phi f_1, \dots, \int \phi f_m)\}$:

GNP Lemma $\Rightarrow \exists K_1, K_2 : \quad \phi: \mathbb{R}^n \rightarrow [0, 1]$

$$\psi^*(t) = \begin{cases} 1 & \text{if } f_T(t|\theta') > K_1 f_T(t|\theta_0) + K_2 t f_T(t|\theta_0) \\ 0 & \text{otherwise} \end{cases}$$

maximizes $\beta_{\psi}(\theta')$ subject to (i) and (ii)

rejection region: $h_T(t) c(\theta') e^{\theta' t} > K_1 h_T(t) c(\theta_0) e^{\theta_0 t} + K_2 h_T(t) c(\theta_0) t e^{\theta_0 t}$

$$\frac{c(\theta')}{c(\theta_0)} e^{\underbrace{(\theta' - \theta_0)}_b t} > K_1 + K_2 t$$

$$\Leftrightarrow e^{bt} > \frac{c(\theta')}{c(\theta_0)} (K_1 + K_2 t) = a_1 + a_2 t$$

$$\Leftrightarrow g(t) := e^{bt} - a_2 t > a_1$$

$$g''(t) = b^2 e^{bt} > 0 \Rightarrow g \text{ is convex}$$

$$\psi^*(t) = \begin{cases} 1 & \text{if } g(t) > a_1 \\ 0 & \text{if } g(t) < a_1 \end{cases}$$

$= \psi_1(t)$ (due to MLR)

case 1: $g(t) \uparrow t$ $\{g(t) > a_1\} = \{t > c\}$ is a non-decreasing fn. of t . So power fn. $\mathbb{E}_\theta \psi_1(T) \uparrow \theta$

$$\text{Then } \beta_{\psi'}(\theta_0) \neq 0 \quad (\Rightarrow \Leftarrow)$$

case 2: $g(t) \downarrow t \quad (\Rightarrow \Leftarrow)$

case 3: $g(t)$ is U-shaped $\{g(t) > a_1\}$
 $= \{t < c_1 \text{ or } t > c_2\}$

$$\text{So } \psi^*(t) = \begin{cases} 1 & \text{if } t < c_1 \text{ or } t > c_2 \\ 0 & c_1 \leq t \leq c_2 \end{cases}$$

$$\text{where } c_1, c_2 : (i) \mathbb{E}_{\theta_0} \psi^*(T) = \alpha$$

$$(ii) \mathbb{E}_{\theta_0}(T \psi^*(T)) = \alpha \mathbb{E}_{\theta_0} T$$

maximizes $\beta_{\psi}(\theta')$

$\therefore \psi^*$ does not depend on θ'

then: $\psi^*(t) = \begin{cases} 1 & \text{if } t < c_1 \text{ or } t > c_2 \\ 0 & \text{if } c_1 \leq t \leq c_2 \end{cases}$ is UMP test satisfying (i), (ii)

$$\psi^* \in B = \{\text{tests satisfying (i), (ii)}\}$$

Since $\psi_0 \equiv \alpha$ satisfy (i) and (ii), compare power of ψ_0 and ψ^*

$$\beta_{\psi^*}(\theta) \geq \beta_{\psi_0}(\theta) \quad \forall \theta \neq \theta_0$$

$\Rightarrow \psi^*$ unbiased size- α

$\Rightarrow \psi^*$ is UMPU size- α test

$$T-a \stackrel{d}{=} -(T-a) \quad \text{then } E_{\theta_0}(\psi^*(T)) = \alpha$$

$$E_{\theta_0}(T\psi^*(T)) = E_{\theta_0}((T-a)\psi^*(T)) + aE_{\theta_0}(\psi^*(T))$$

$$= 0 \quad ?? \quad = 0$$